

tion-3+ described in the previous section.

Intensified CCD

Intensified CCD, or ICCD, successfully exploits the optical amplification provided by an image intensifier to overcome the limitations of the basic CCD sensor. Two important features of an ICCD are high optical gain and gateable operation. Both attributes are characteristic of the image intensifier tube.

Though image intensifiers were initially developed for the military and law enforcement agencies for a range of deployment scenarios in surveillance, targeting and navigation, development of ICCD technology and devices extended their usage to many a scientific application in spectroscopy, scientific and industrial imaging, and medical diagnostics. In fact, development of image intensifier tubes is increasingly being driven by scientific applications.

Construction. An ICCD primarily comprises an image intensifier tube whose light output is coupled with a CCD sensor. Output of the image intensifier is coupled with the CCD, typically by a fibre-optic coupler (Fig. 4). A fibre-coupled system is physically compact with low optical distortion levels. High efficiency fibre-optic coupling also allows the image intensifier tube to operate at lower gains, which, in turn, results in better dynamic range performance.

A lens-coupled ICCD uses a lens between the output of the image intensifier and the CCD (Fig. 5). Lens coupling offers flexibility of using the ICCD sensor in non-intensified mode by allowing the image intensifier to be removed.

Disadvantages of lens-coupled ICCDs are larger physical size, lower coupling efficiencies and increased scatter. Power supply is another important constituent part of the ICCD sensor. The power supply section generates DC voltage (typically 600 to 900 volts) for the MCP to achieve desired gain, DC voltage (typically 4kV to 8kV) for the phosphor screen and voltage pulses (typically 200 volts) for gated operation of the photocathode.

The gating pulse width and rise/

fall time depends on desired gating parameters. Gating pulse width of less than a nanosecond and rise/fall time of a small fraction of a nanosecond are achievable.

Characteristic features. Important characteristic features of an ICCD include spectral response, spatial resolution, gating time and repetition rate, noise, sensitivity, dynamic range and frame rate. The spectral response of an ICCD camera is primarily determined by the input window, photocathode materials and photocathode size used in the image intensifier tube.

The noise and, hence, sensitivity of the ICCD is also governed by the image intensifier. A noise component called dark current component, also known as effective background illumination (EBI), originates from thermally-generated charge in the photocathode. Dark current is generally not an issue when using short gate times.

Spatial resolution is defined as the number of line pairs a camera can resolve per millimetre. Among the several methods available for measuring the resolution of an optical system, modulation transfer function (MTF) is in common use.

Dynamic range of the ICCD is governed by the CCD section and varies inversely with gain of the ICCD. A higher dynamic range CCD used in the ICCD will result in a higher dynamic range ICCD camera. As gain increases, smaller signals that can be accommodated are compensated by the lower read noise to keep the dynamic range constant.

When the read noise drops below a single photon level, dynamic range of the ICCD starts dropping as gain increases further. Frame rates of an

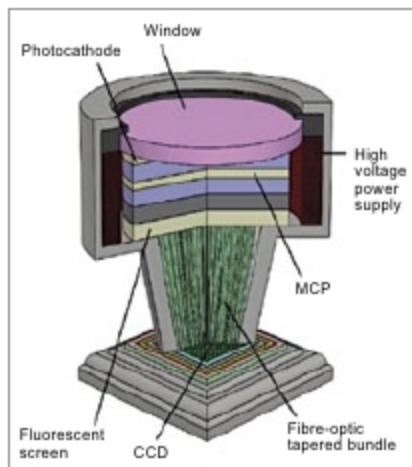


Fig. 4: Construction of ICCD with fibre-optic output coupling

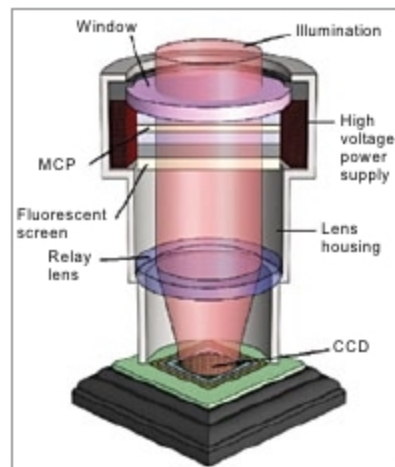


Fig. 5: Construction of ICCD with lens coupling

ICCD are governed by CCD specifications, especially the number of pixels and pixel readout rate.

Thermal imaging

CCD and CMOS sensors have a spectral response sensitive to the visible region of the electromagnetic spectrum, with the trailing part extending slightly to near IR region.

A thermal imaging sensor, on the other hand, uses focal plane arrays comprising IR sensing elements that respond to mid (3 to 5 microns) and long IR (8 to 14 microns) regions. IR energy radiated by the object to be imaged is incident on the thermal imaging sensor, which uses a series of complex mathematical algorithms to construct the image visible to the viewer. As the thermal imaging sensor does not need visible light for operation, it can see in total darkness.

Thermal imaging sensors are far more expensive than their visible spectrum counterparts. In view of their military applications, high-end devices are often export-restricted.

Operational basics. A thermal imaging sensor makes use of thermal radiation emitted by the target or scene of interest to generate its image. Essentially, it comprises front-end optical system, two-dimensional array of IR detector elements and image processing circuitry to produce output in the desired format. The front-end optical system focuses IR radiation emitted by all objects in view on a two-dimensional array of IR detector elements, which create a detailed temperature pattern of it, called a thermogram.

The thermogram is generated from several thousand points in the field-of-view of the detector array. The thermal imager measures very small relative temperature differences and converts otherwise invisible heat patterns into clear, visible images that can be seen through either a viewfinder or a monitor.

Most thermal imaging sensors scan at a rate of 30 times a second. These can sense temperatures in the range of -20°C to $+2000^{\circ}\text{C}$, and can sense temperature changes as small as 0.1°C . In the next step, temperature pattern is translated into electronic impulses. The signal processing unit converts these